

DS-02

Mathematical Formulation of DSteele

**A 3D Spectral-Element / Finite-Layer Solution for Pavement Response under
Moving Loads**

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DS-02 • Technical Article

Purpose and Scope

DS-01 introduced DSteele from the perspective of its development, motivation, implementation, and application to moving-load and Traffic Speed Deflection Device (TSDD) analysis. This companion article documents the mathematical formulation underlying the model. The derivation that follows is retained from my original, unpublished DSteele formulation manuscript with only the surrounding context reorganized for this article.

The purpose of DS-02 is not to reproduce the DSteele source code. Rather, it is to document the mechanics and mathematical structure of the solution: the governing equations, transformation to the wavenumber domain, spectral-element stiffness formulation for finite layers and the semi-infinite halfspace, treatment of elastic and viscoelastic properties, global assembly, moving surface loading, and inverse-transformation of the response back to the physical domain.

Mathematical Development of DSteele

As indicated in one of my previous LinkedIn posts (<https://lnkd.in/p/eWKApk64>), I derived the mathematical derivation of DSteele by hand. The hand-written derivation extended across more than twenty pages and included the displacement-potential solution, evaluation of displacement and stress components at layer boundaries that were ultimately used to construct the spectral-element stiffness matrix.

Development of the formulation was iterative. Because an error discovered late in the derivation could propagate into the layer stiffness matrices, portions of the derivation were restarted and checked repeatedly before being translated into code. The photographs below preserve part of that original development record. They are included as technical provenance for the formulation, not as a substitute for the mathematical development that follows.

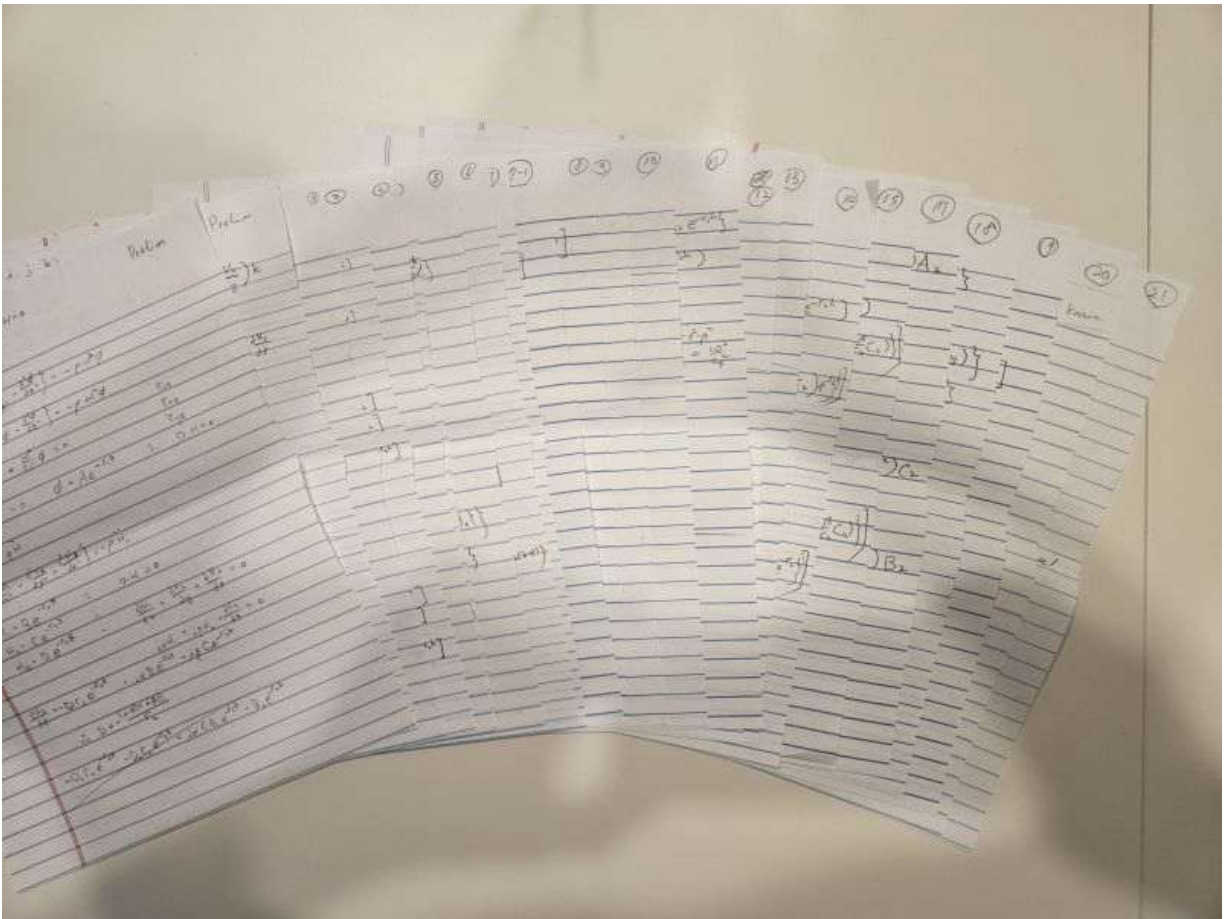


Figure 1. Original handwritten derivation developed during the formulation of DSteele. The derivation progressed from the governing equations and displacement potentials through finite-layer displacement and stress relationships used to construct the spectral-element stiffness matrices.

Elements of $[L] = [L'] [A]$

① $z=0$

$$u_z = i\alpha A_1 + i\alpha e^{-r_1 h} A_2 - \frac{\alpha\beta}{r_1} B_1 + \frac{\alpha\beta}{r_1} e^{-r_1 h} B_2 + (r_2 - \frac{\beta^2}{r_2}) C_1 + (-r_2 e^{-r_2 h} + \frac{\beta^2}{r_2} e^{-r_2 h}) C_2$$

$$u_y = i\beta A_1 + i\beta e^{-r_1 h} A_2 + (-r_2 + \frac{\beta^2}{r_2}) B_1 + (r_2 e^{-r_2 h} - \frac{\beta^2}{r_2} e^{-r_2 h}) B_2 + \frac{\alpha\beta}{r_1} C_1 - \frac{\alpha\beta}{r_2} e^{-r_2 h} C_2$$

$$u_x = -r_1 A_1 + r_1 e^{-r_1 h} A_2 - i\beta B_1 - i\beta e^{-r_1 h} B_2 + i\alpha C_1 + i\alpha e^{-r_1 h} C_2$$

② $z=h$

$$u_z = i\alpha e^{-r_1 h} A_1 + i\alpha A_2 - \frac{\alpha\beta}{r_1} e^{-r_1 h} B_1 + \frac{\alpha\beta}{r_1} B_2 + (r_2 e^{-r_2 h} - \frac{\beta^2}{r_2} e^{-r_2 h}) C_1 + (-r_2 + \frac{\beta^2}{r_2}) C_2$$

$$u_y = i\beta e^{-r_1 h} A_1 + i\beta A_2 + (-r_2 e^{-r_2 h} + \frac{\beta^2}{r_2} e^{-r_2 h}) B_1 + (r_2 - \frac{\beta^2}{r_2}) B_2 + \frac{\alpha\beta}{r_1} e^{-r_1 h} C_1 - \frac{\alpha\beta}{r_2} C_2$$

Figure 2. Close-up from the original handwritten development of the finite-layer displacement relationships, including evaluation of displacement components at the upper and lower layer boundaries prior to their organization in matrix form.

Detailed DSteele Formulation

The formulation proceeds from the three-dimensional equations of motion for an elastic, homogeneous continuum. Helmholtz decomposition is then used to express the displacement field in terms of scalar and vector potentials. For a load moving at constant speed, the solution is written in a two-dimensional Fourier representation so that derivatives with respect to space and time can be handled algebraically in the transformed domain.

The resulting harmonic solutions are used to derive stiffness matrices for two-noded finite-thickness layers and a one-noded semi-infinite element. Elastic or viscoelastic layer properties are incorporated into those element relationships, after which the individual elements are assembled into the global

pavement system. The transformed nodal response is solved for each wavenumber pair and subsequently transformed back to obtain the pavement response associated with the moving load. Details of DSteele formulation is provided in the following.

FORMULATION OF DSTEELE

GOVERNING EQUATIONS

The formulation of DSteele starts with the classical equation of motion for an elastic, homogeneous continuum without body forces. In terms of the displacement vector, $\mathbf{u}$, the equation is given as the following.

$$(\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u} = \rho \ddot{\mathbf{u}} \quad (1)$$

where λ and μ are the lamé constants, and ρ is the mass density of the material. The stresses of an isotropic material satisfying the governing equation shown above, are given by the generalized Hooke's law written as the following.

$$\boldsymbol{\sigma} = \lambda \text{tr}(\boldsymbol{\varepsilon}) \mathbf{I} + 2\mu \boldsymbol{\varepsilon} \quad (2)$$

where $\mathbf{I}$ is the identity tensor and $\boldsymbol{\varepsilon}$ is the strain tensor given as the following.

$$\boldsymbol{\varepsilon} = \frac{1}{2} \left\{ \nabla \mathbf{u} + (\nabla \mathbf{u})^T \right\} \quad (3)$$

In a cartesian coordinate system such as the one shown in Figure 3, Equation (1) represents a complicated system of partial differential equations that need to be solved for three components of the displacement vector (i.e., u_x , u_y , u_z). As pointed out by Graff (1991), Equation (1) can be decoupled into a more simpler form by means of the Helmholtz decomposition given as the following.

$$\mathbf{u} = \nabla \Phi + \nabla \times \mathbf{H} \quad (4)$$

where Φ represent a scalar potential and $\mathbf{H}$ is a vector potential whose divergence vanishes. In other words, the following condition must hold for the vector potential $\mathbf{H}$.

$$\nabla \cdot \mathbf{H} = 0 \quad (5)$$

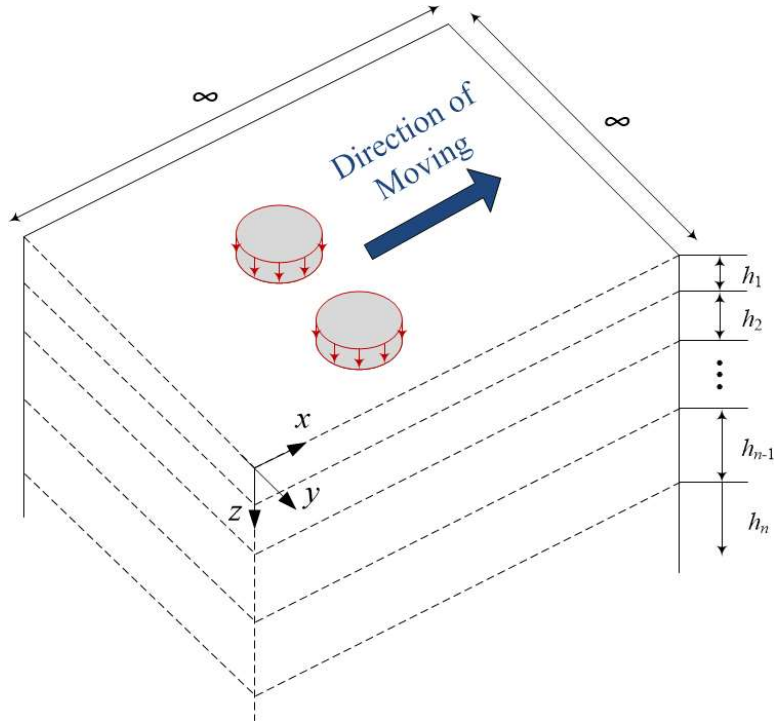


Figure 3. Cartesian Coordinate System with Multiple Layers on a Halfspace

Substituting Equation (4) into Equation (1), one arrives at the following decoupled equations for Φ and $\mathbf{H}$.

$$(\lambda + 2\mu)\nabla^2\Phi = \rho\ddot{\Phi} \quad (6)$$

$$\mu\nabla^2\mathbf{H} = \rho\ddot{\mathbf{H}} \quad (7)$$

Similar to the solutions provided by Siddharthan et. al. (1998, 2000) and Nielsen (2019), the solutions of the above equations under a load moving at a constant speed, V , can be written in the form of an inverse Fourier Transform. More specifically, the solutions for the scalar and the vector potentials are written as the following.

$$\Phi(x-Vt, y, z) = \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \hat{\Phi}(k_x, k_y, z) e^{ik_x(x-Vt)} e^{ik_y y} dk_x dk_y \quad (8)$$

$$\mathbf{H}(x-Vt, y, z) = \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \hat{\mathbf{H}}(k_x, k_y, z) e^{ik_x(x-Vt)} e^{ik_y y} dk_x dk_y \quad (9)$$

where $i = \sqrt{-1}$, $\hat{f}(k_x, k_y, z)$ is the 2-dimensional (2D) Fourier transform of the function $f(x, y, z)$ along the x and y axes, and k_x and k_y are the wave numbers along their respective axes.

The primary advantage of writing the solutions in the form of Equations (8) and (9) is that the partial derivatives with respect to x , y , and t reduce to simple algebraic multiplications. For example, by denoting the Fourier transform pairs as $f \leftrightarrow \hat{f}$, it is easily seen from Equation (8) that the partial derivative of Φ

with respect to x , is obtained as $\partial\Phi/\partial x \leftrightarrow ik_x\hat{\Phi}$. Therefore, the solution of the above equations can be obtained more conveniently in the transformed domain (i.e., wave number or frequency domain).

In the frequency domain, the components of the displacement vector $\mathbf{u}$ can be written as the following in terms of the potentials.

$$\hat{u}_x = ik_x\hat{\Phi} + ik_y\hat{H}_z - \frac{\partial\hat{H}_y}{\partial z} \quad (10)$$

$$\hat{u}_y = ik_y\hat{\Phi} + \frac{\partial\hat{H}_x}{\partial z} - ik_x\hat{H}_z \quad (11)$$

$$\hat{u}_z = \frac{\partial\hat{\Phi}}{\partial z} + ik_x\hat{H}_y - ik_y\hat{H}_x \quad (12)$$

where $H_x \leftrightarrow \hat{H}_x$, $H_y \leftrightarrow \hat{H}_y$, and $H_z \leftrightarrow \hat{H}_z$ are the x , y , and z components of the vector potential $\mathbf{H}$, respectively.

SOLUTIONS FOR HARMONIC POTENTIALS

In order to solve for the harmonic components of $\hat{\Phi}$ and $\hat{\mathbf{H}}$ in Equations (8) and (9), it is first necessary to transform Equations (6) and (7) into the same wave number or frequency domain. For the scalar potential Φ , transforming Equation (6) into the frequency domain results in the following.

$$(\lambda + 2\mu) \left\{ -k_x^2\hat{\Phi} - k_y^2\hat{\Phi} + \frac{\partial^2\hat{\Phi}}{\partial z^2} \right\} = -\rho\omega^2\hat{\Phi} \quad (13)$$

where $\omega = -k_x V$. The above equation can be rewritten in the form of a classical wave equation as:

$$\frac{\partial^2\hat{\Phi}}{\partial z^2} - r_1^2\hat{\Phi} = 0 \quad (14)$$

where

$$r_1 = k_x^2 + k_y^2 - \left(\frac{\omega}{c_1} \right)^2 \quad (15)$$

and

$$c_1 = \sqrt{\frac{\lambda + 2\mu}{\rho}} \quad (16)$$

After dropping the term corresponding to the stress wave propagating in the negative z axis, the solution for the scalar potential is obtained as the following (Ewing et. al. 1957; Graff 1991).

$$\hat{\Phi} = A_1 e^{-r_1 z} \quad (17)$$

where A_1 is an arbitrary constant.

Similarly, one arrives at the following wave equation for the vector potential $\mathbf{H}$, by following the same procedure shown above.

$$\frac{\partial^2 \hat{\mathbf{H}}}{\partial z^2} - r_2^2 \hat{\mathbf{H}} = 0 \quad (18)$$

where

$$r_2 = k_x^2 + k_y^2 - \left(\frac{\omega}{c_2} \right)^2 \quad (19)$$

and

$$c_2 = \sqrt{\frac{\mu}{\rho}} \quad (20)$$

The solution for each component of $\mathbf{H}$ is obtained as the following, after dropping the terms that result in the unbounded results.

$$\hat{H}_x = B_1 e^{-r_2 z} \quad (21-1)$$

$$\hat{H}_y = C_1 e^{-r_2 z} \quad (21-2)$$

$$\hat{H}_z = D_1 e^{-r_2 z} \quad (21-3)$$

where B_1 , C_1 , and D_1 are arbitrary constants.

Stiffness Matrix of a Two-Noded Element with Finite Thickness

The solution for the scalar potential shown in Equation (17) and those for the vector potential shown in Equation (21) only account for the stress waves propagating in the positive z -direction (i.e., downwards, see Figure 3). However, for a layer with finite thickness, h , the stress wave not only propagates downwards, but also upwards due to the reflection and refraction at the lower boundary. To account for the waves propagating upwards from the lower boundary, another term is added to the right-hand side of Equations (18) and (21) (Al-Khoury et. al., 2001; Lee, 2013a; Lee, 2013b). Thus, the solution for the scalar potential is written as the following.

$$\hat{\Phi} = A_1 e^{-r_1 z} + A_2 e^{-r_1 (h-z)} \quad (22)$$

where A_1 and A_2 are arbitrary constants. Similarly, the solution for the vector potential is written as:

$$\hat{H}_x = B_1 e^{-r_2 z} + B_2 e^{-r_2 (h-z)} \quad (23-1)$$

$$\hat{H}_y = C_1 e^{-r_2 z} + C_2 e^{-r_2(h-z)} \quad (23-2)$$

$$\hat{H}_z = D_1 e^{-r_2 z} + D_2 e^{-r_2(h-z)} \quad (23-3)$$

where B_j , C_j , and D_j ($j = 1, 2$) are arbitrary constants. Substituting the above equation into the divergence condition of $\mathbf{H}$, i.e., Equation (5), one arrives at the following expression for the arbitrary constants of $\hat{H}_z$.

$$D_1 = \frac{ik_x}{r_2} B_1 + \frac{ik_y}{r_2} C_1 \quad (24-1)$$

$$D_2 = -\frac{ik_x}{r_2} B_2 - \frac{ik_y}{r_2} C_2 \quad (24-2)$$

By substituting Equations (22) through (24) into Equations (10) through (12), the displacements are obtained as the following.

$$\begin{aligned} \hat{u}_x = & ik_x \left\{ A_1 e^{-r_1 z} + A_2 e^{-r_1(h-z)} \right\} - K_{xy} \left\{ B_1 e^{-r_2 z} - B_2 e^{-r_2(h-z)} \right\} \\ & - K_y \left\{ C_1 e^{-r_2 z} - C_2 e^{-r_2(h-z)} \right\} \end{aligned} \quad (25)$$

$$\begin{aligned} \hat{u}_y = & ik_y \left\{ A_1 e^{-r_1 z} + A_2 e^{-r_1(h-z)} \right\} + K_x \left\{ B_1 e^{-r_2 z} - B_2 e^{-r_2(h-z)} \right\} \\ & + iK_{xy} \left\{ C_1 e^{-r_2 z} - C_2 e^{-r_2(h-z)} \right\} \end{aligned} \quad (26)$$

$$\begin{aligned} \hat{u}_z = & -r \left\{ A_1 e^{-r_1 z} - A_2 e^{-r_1(h-z)} \right\} - ik_y \left\{ B_1 e^{-r_2 z} + B_2 e^{-r_2(h-z)} \right\} \\ & + ik_x \left\{ C_1 e^{-r_2 z} + C_2 e^{-r_2(h-z)} \right\} \end{aligned} \quad (27)$$

where

$$K_{xy} = \frac{k_x k_y}{r_2} \quad (28)$$

$$K_x = \frac{k_x^2}{r_2} - r_2 \quad (29)$$

$$K_y = \frac{k_y^2}{r_2} - r_2 \quad (30)$$

The equations for the relevant stresses are obtained by substituting Equation (3) into Equation (2), then taking the Fourier transform with respect to x and y . Substituting Equations (25) through (30) into the resulting equations yields the following.

$$\begin{aligned}\hat{\tau}_{zx} &= \mu \left(\frac{\partial \hat{u}_x}{\partial z} + ik_x \hat{u}_z \right) \\ &= -2i\mu k_x r_1 \left\{ A_1 e^{-r_1 z} - A_2 e^{-r_1 (h-z)} \right\} + 2\mu k_x k_y \left\{ B_1 e^{-r_2 z} + B_2 e^{-r_2 (h-z)} \right\} \\ &\quad + \Gamma_3 \left\{ C_1 e^{-r_2 z} + C_2 e^{-r_2 (h-z)} \right\}\end{aligned}\quad (31)$$

$$\begin{aligned}\hat{\tau}_{yz} &= \mu \left(\frac{\partial \hat{u}_y}{\partial z} + ik_y \hat{u}_z \right) \\ &= -2i\mu k_y r_1 \left\{ A_1 e^{-r_1 z} - A_2 e^{-r_1 (h-z)} \right\} + \Gamma_2 \left\{ B_1 e^{-r_2 z} + B_2 e^{-r_2 (h-z)} \right\} \\ &\quad - 2\mu k_x k_y \left\{ C_1 e^{-r_2 z} + C_2 e^{-r_2 (h-z)} \right\}\end{aligned}\quad (32)$$

$$\begin{aligned}\hat{\sigma}_z &= \lambda \left(ik_x \hat{u}_x + ik_y \hat{u}_y + \frac{\partial \hat{u}_z}{\partial z} \right) + 2\mu \frac{\partial \hat{u}_z}{\partial z} \\ &= \Gamma_1 \left\{ A_1 e^{-r_1 z} + A_2 e^{-r_1 (h-z)} \right\} + 2i\mu k_y r_2 \left\{ B_1 e^{-r_2 z} - B_2 e^{-r_2 (h-z)} \right\} \\ &\quad - 2i\mu k_x r_2 \left\{ C_1 e^{-r_2 z} - C_2 e^{-r_2 (h-z)} \right\}\end{aligned}\quad (33)$$

where

$$\Gamma_1 = \lambda \left(r_1^2 - k_x^2 - k_y^2 \right) + 2\mu r_1^2 \quad (34)$$

$$\Gamma_2 = \mu \left(k_y^2 - k_x^2 + r_2^2 \right) \quad (35)$$

$$\Gamma_3 = \mu \left(k_y^2 - k_x^2 - r_2^2 \right) \quad (36)$$

To construct the stiffness matrix of a layer with finite thickness, h , it is necessary to obtain the displacements and the stresses at the upper and lower boundaries. For the displacements, this is achieved easily by substituting $z = 0$ for the upper boundary and $z = h$ for the lower boundary in Equations (25) through (27). The resulting displacements are obtained as the following in matrix form.

$$\hat{\mathbf{u}} = \mathbf{S}_1 \cdot \mathbf{A} \quad (37)$$

where $\hat{\mathbf{u}} = \{\hat{u}_{x1} \ \hat{u}_{y1} \ \hat{u}_{z1} \ \hat{u}_{x2} \ \hat{u}_{y2} \ \hat{u}_{z2}\}^T$ and $\mathbf{A} = \{A_1 \ A_2 \ B_1 \ B_2 \ C_1 \ C_2\}^T$. $\hat{u}_{x1}$, $\hat{u}_{y1}$, and $\hat{u}_{z1}$ are the displacements at the upper boundary along the x, y, and z axes, respectively, and $\hat{u}_{x2}$, $\hat{u}_{y2}$, and $\hat{u}_{z2}$ are their counterparts at the lower boundary. The matrix $\mathbf{S}_1$ is given as:

$$\mathbf{S}_1 = \begin{bmatrix} ik_x & ik_x Q_1 & -K_{xy} & K_{xy} Q_2 & -K_y & K_y Q_2 \\ ik_y & ik_y Q_1 & K_x & -K_x Q_2 & K_{xy} & -K_{xy} Q_2 \\ -r_1 & r_1 Q_1 & -ik_y & -ik_y Q_2 & ik_x & ik_x Q_2 \\ ik_x Q_1 & ik_x & -K_{xy} Q_2 & K_{xy} & -K_y Q_2 & K_y \\ ik_y Q_1 & ik_y & K_x Q_2 & -K_x & K_{xy} Q_2 & -K_{xy} \\ -r_1 Q_1 & r_1 & -ik_y Q_2 & -ik_y & ik_x Q_2 & ik_x \end{bmatrix} \quad (38)$$

where

$$Q_1 = e^{-r_1 h} \quad (39)$$

$$Q_2 = e^{-r_2 h} \quad (40)$$

Similarly, substituting $z = 0$ for the upper boundary and $z = h$ for the lower boundary in Equations (31) through (33) yields the following for the stresses in matrix form.

$$\hat{\boldsymbol{\sigma}} = \mathbf{S}_2 \cdot \mathbf{A} \quad (41)$$

where $\hat{\boldsymbol{\sigma}} = \{\hat{\tau}_{zx1} \ \hat{\tau}_{yz1} \ \hat{\sigma}_{z1} \ \hat{\tau}_{zx2} \ \hat{\tau}_{yz2} \ \hat{\sigma}_{z2}\}^T$ in which the subscripts 1 and 2 indicate the corresponding stresses at the upper and lower boundaries, respectively. The matrix $\mathbf{S}_2$ is given as:

$$\mathbf{S}_2 = \begin{bmatrix} -2i\mu k_x r_1 & 2i\mu k_x r_1 Q_1 & 2\mu k_x k_y & 2\mu k_x k_y Q_2 & \Gamma_3 & \Gamma_3 Q_2 \\ -2i\mu k_y r_1 & 2i\mu k_y r_1 Q_1 & \Gamma_2 & \Gamma_2 Q_2 & -2\mu k_x k_y & -2\mu k_x k_y Q_2 \\ \Gamma_1 & \Gamma_1 Q_1 & 2i\mu k_y r_2 & -2i\mu k_y r_2 Q_2 & -2i\mu k_x r_2 & 2i\mu k_x r_2 Q_2 \\ -2i\mu k_x r_1 Q_1 & 2i\mu k_x r_1 & 2\mu k_x k_y Q_2 & 2\mu k_x k_y & \Gamma_3 Q_2 & \Gamma_3 \\ -2i\mu k_y r_1 Q_1 & 2i\mu k_y r_1 & \Gamma_2 Q_2 & \Gamma_2 & -2\mu k_x k_y Q_2 & -2\mu k_x k_y \\ \Gamma_1 Q_1 & \Gamma_1 & 2i\mu k_y r_2 Q_2 & -2i\mu k_y r_2 & -2i\mu k_x r_2 Q_2 & 2i\mu k_x r_2 \end{bmatrix} \quad (42)$$

With the matrices $\mathbf{S}_1$ and $\mathbf{S}_2$ defined as the above, the stiffness matrix of a layer with finite thickness (i.e., the stiffness matrix of a two-noded element, with one node at the upper boundary and the other at the lower boundary) is obtained by relating the nodal displacements to boundary tractions. In other words, the stiffness matrix is obtained from:

$$\begin{Bmatrix} \hat{T}_{zx1} \\ \hat{T}_{yz1} \\ \hat{T}_{z1} \\ \hat{T}_{zx2} \\ \hat{T}_{yz2} \\ \hat{T}_{z2} \end{Bmatrix} = \begin{Bmatrix} -\hat{\tau}_{zx1} \\ -\hat{\tau}_{yz1} \\ -\hat{\sigma}_{z1} \\ \hat{\tau}_{zx2} \\ \hat{\tau}_{yz2} \\ \hat{\sigma}_{z2} \end{Bmatrix} = \mathbf{S}_{2\text{-noded}} \cdot \begin{Bmatrix} \hat{u}_{x1} \\ \hat{u}_{y1} \\ \hat{u}_{z1} \\ \hat{u}_{x2} \\ \hat{u}_{y2} \\ \hat{u}_{z2} \end{Bmatrix} \quad (43)$$

in which the stiffness matrix, $\mathbf{S}_{2\text{-noded}}$, is given as the following.

$$\mathbf{S}_{2\text{-noded}} = \mathbf{D} \cdot \mathbf{S}_2 \cdot \mathbf{S}_1^{-1} \quad (44)$$

where

$$\mathbf{D} = \text{diag} \{-1 \quad -1 \quad -1 \quad 1 \quad 1 \quad 1\}^T \quad (45)$$

Stiffness Matrix of a One-Noded, Semi-Infinite Element

An example of the one-noded, semi-infinite element (i.e., a halfspace element) was schematically shown as the bottom layer in Figure 3. Obviously, the one-noded element only has a single boundary at the top of the layer. In other words, this element does not have a lower boundary from which the stress waves can reflect and refract back to the upper boundary. Therefore, the stiffness matrix of a semi-infinite element can be obtained in the same manner as the two-noded element without the terms that govern the stress waves propagating upwards [i.e., $A_2 = B_2 = C_2 = D_2 = 0$ in Equations (22) and (23)].

Following the same mathematical procedure previously shown for the two-noded element, one arrives at the following for the stiffness matrix of a one-noded element.

$$\mathbf{S}_{1\text{-noded}} = -\mathbf{S}_3 \cdot \mathbf{S}_4^{-1} \quad (46)$$

where the matrices $\mathbf{S}_3$ and $\mathbf{S}_4$ are obtained from the displacement and stress functions as shown below.

$$\mathbf{S}_3 = \begin{bmatrix} ik_x & -K_{xy} & -K_y \\ ik_y & K_x & K_{xy} \\ -r & -ik_y & ik_x \end{bmatrix} \quad (47)$$

$$\mathbf{S}_4 = \begin{bmatrix} -2i\mu k_x r & 2\mu k_x k_y & \Gamma_3 \\ -2i\mu k_y r & \Gamma_2 & -2\mu k_x k_y \\ \Gamma_1 & 2i\mu k_y r_2 & -2i\mu k_x r_2 \end{bmatrix} \quad (48)$$

Incorporating Elastic and Viscoelastic Layer Properties

For an elastic material whose stiffness characteristics are time-independent, the lamé constants can easily be obtained from the Young's modulus, E , and the Poisson's ratio, ν . From the classical theory of elasticity, these relationships are given as:

$$\mu = \frac{E}{2(1 + \nu)} \quad (49)$$

$$\lambda = \frac{E\nu}{(1 + \nu)(1 - 2\nu)} \quad (50)$$

However, as indicated Rizzi (1989) who introduced the spectral element solution approach for layered media, it is recommended that a small amount of damping be added to the lamé constants, since no realistic material is purely elastic. In classical soil dynamics, this is achieved by adding an imaginary, hysteretic damping to the material functions (Siddharthan et. al. 1998, 2000). In other words, the complex lamé constants with damping are given as the following:

$$\mu^* = \mu(1 + 2i\zeta) \quad (51)$$

$$\lambda^* = \lambda(1 + 2i\zeta) \quad (52)$$

where ζ is a damping constant.

The modulus of a viscoelastic material such as Asphalt Concrete (AC) is time (or equivalently, frequency) dependent. A well-known model for the time-dependent viscoelastic modulus is given in terms of a sigmoidal function of the following form.

$$\log E(t) = c_1 + \frac{c_2}{1 + e^{c_3 + c_4 \log(t)}} \quad (53)$$

where $E(t)$ is the time dependent, viscoelastic relaxation modulus at time t , and c_1 through c_4 are model constants. Similarly, the frequency-dependent viscoelastic modulus model is often written as the following.

$$\log(|E^*|) = d_1 + \frac{d_2}{1 + e^{d_3 + d_4 \log(f)}} \quad (54)$$

where $|E^*|$ is the dynamic modulus which is simply the amplitude of the complex modulus $E^* = E' + iE''$, f is the frequency, and d_1 through d_4 are model constants.

Although Equation (54) may seem more natural and adequate than Equation (53) for use with the complex stiffness matrices previously derived in the frequency domain, one drawback of Equation (54) is that the real and imaginary parts of the complex modulus (E' and E'' , respectively) needed for these stiffness matrices cannot be obtained without the knowledge of phase angle at different frequencies (note that Equation (54) does not include any phase angle information). A feasible workaround for finding E' and E'' from Equation (54) is through the Prony series representation of the complex modulus in which E' and E'' are written as the following.

$$E'(\omega) = E_0 + \sum_{i=1}^M \frac{\omega^2 \rho_i^2 E_i}{\omega^2 \rho_i^2 + 1} \quad (55-1)$$

$$E''(\omega) = \sum_{i=1}^M \frac{\omega \rho_i E_i}{\omega^2 \rho_i^2 + 1} \quad (55-2)$$

where E_0 and E_i are the Prony coefficients, M is the number of Prony elements, and ρ_i is the relaxation time for the i^{th} Prony element. From the above, the frequency dependent dynamic modulus can be written as:

$$|E^*(\omega)| = \sqrt{\left\{ E_0 + \sum_{i=1}^M \frac{\omega^2 \rho_i^2 E_i}{\omega^2 \rho_i^2 + 1} \right\}^2 + \left\{ \sum_{i=1}^M \frac{\omega \rho_i E_i}{\omega^2 \rho_i^2 + 1} \right\}^2} \quad (56)$$

To determine the Prony coefficients, it is necessary to fit the dynamic modulus expressed in the form of Equation (56) to those obtained from Equation (54). However, due to the complicated nature of Equation (56), this process involves a Nonlinear Least Squares (NLS) approach which requires iteration over the objective function (which minimizes the error between the $|E^*|$ values from Equation (56) and those from Equation (54)), starting from a reasonable set of seed values (i.e., initial guess) for the Prony coefficients (Lee, 2022).

However, the above NLS process was not desired for use with the spectral element solution developed in this study, primarily for two reasons. First, the $|E^*|$ obtained in the form of Equation (56) depends on the seed values, i.e., using a different set of seed values may result in different $|E^*|$ values which in turn, may result in different responses obtained from the spectral element solution. Second, the NLS requires iteration which increases the runtime of the implemented algorithm, and the increase in runtime is not expected to be consistent (as the NLS may need different number of iterations before converging, depending on the seed values).

Due to the above challenges with determining the Prony coefficients from $|E^*|$, an alternate method was adopted for implementation. Rather than finding the Prony coefficients from the frequency-dependent functions [i.e., Equations (54) and (56)], these coefficients were determined by fitting the Prony series representation in time-domain to the relaxation modulus represented in Equation (53). The time-domain Prony series model for the relaxation modulus is given as the following.

$$E(t) = E_0 + \sum_{i=1}^M E_i \cdot e^{-\frac{t}{\rho_i}} \quad (57)$$

Obtaining the Prony coefficients from Equation (57) is carried out by fitting the $E(t)$ in this equation to those obtained from Equation (53). Due to the simple form of Equation (57), this fitting process can be achieved through a Linear Least Squares (LLS) rather than a NLS. More specifically, the LLS reduces to a set of linear algebraic equations that can be written as the following in matrix form (See Lee, 2022 for a detailed derivation and proof).

$$\{E_0 \quad E_1 \quad \cdots \quad E_M\}^T = \mathbf{M}_{M \times M}^{-1} \cdot \mathbf{B}_{M \times 1} \quad (58)$$

where

$$\mathbf{M}_{M \times M} = \begin{bmatrix} n & \sum_k e^{-\frac{t_k}{\rho_1}} & \dots & \sum_k e^{-\frac{t_k}{\rho_M}} \\ \sum_k e^{-\frac{t_k}{\rho_1}} & \sum_k e^{-\frac{t_k}{\rho_1}} \cdot e^{-\frac{t_k}{\rho_1}} & & \sum_k e^{-\frac{t_k}{\rho_1}} \cdot e^{-\frac{t_k}{\rho_M}} \\ \vdots & & \ddots & \vdots \\ \sum_k e^{-\frac{t_k}{\rho_M}} & \sum_k e^{-\frac{t_k}{\rho_M}} \cdot e^{-\frac{t_k}{\rho_1}} & \dots & \sum_k e^{-\frac{t_k}{\rho_M}} \cdot e^{-\frac{t_k}{\rho_M}} \end{bmatrix} \quad (59)$$

$$\mathbf{B}_{M \times 1} = \left\{ \sum_k E_k \quad \sum_k E_k e^{-\frac{t_k}{\rho_1}} \quad \dots \quad \sum_k E_k e^{-\frac{t_k}{\rho_M}} \right\}^T \quad (60)$$

and t_k ($k = 0, 1, 2, \dots, n$) is the discrete time values at which the value of $E(t)$ is obtained from Equations (53) and (57) for LLS. As seen from Equation (58), the LLS does not require any iterations (i.e., it is solved by inverting the matrix $\mathbf{M}$ once) and does not require nor depend on any seed values (i.e., initial guess).

Once the Prony coefficients are obtained from Equation (58), the real and imaginary parts of the complex modulus (E' and E'' , respectively) needed for the spectral element solution can easily be obtained from Equation (55).

Global Stiffness Matrix and Nodal Displacements

Using the element stiffness matrices derived previously, the global stiffness matrix, $\mathbf{S}_{\text{Global}}$, is constructed in a manner similar to the Finite Element Method (FEM) (Al-Khoury et. al., 2001; Lee, 2013a, 2013b). The global stiffness matrix is a $(3 \cdot n + 3)$ by $(3 \cdot n + 3)$ square matrix (n = total number of layers) whose constructional schematics are shown in Figure 4(a) for pavements with a halfspace and in Figure 4(b) for those with a shallow bedrock. In Figure 4, any region that is overlapped by two distinct element stiffness matrices indicate the components of the respective element matrices are summed. Additional details regarding the global stiffness matrix construction can be found in Al-Khoury et. al. (2001) and Lee (2013a, 2013b), and shall not be repeated herein.

$$\mathbf{S}_{\text{Global}} = \begin{bmatrix} \begin{bmatrix} \vdots \\ \vdots \end{bmatrix} & \begin{bmatrix} \mathbf{S}_{2\text{-noded}}^1 \\ \vdots \end{bmatrix} \\ \begin{bmatrix} \vdots \\ \vdots \end{bmatrix} & \begin{bmatrix} \mathbf{S}_{2\text{-noded}}^2 \\ \vdots \end{bmatrix} \\ \vdots & \vdots \\ \begin{bmatrix} \vdots \\ \vdots \end{bmatrix} & \begin{bmatrix} \mathbf{S}_{2\text{-noded}}^{n-1} \\ \mathbf{S}_{1\text{-noded}}^n \end{bmatrix} \end{bmatrix}$$

(a)

$$\mathbf{S}_{\text{Global}} = \begin{bmatrix} \begin{bmatrix} \vdots \\ \vdots \end{bmatrix} & \begin{bmatrix} \mathbf{S}_{2\text{-noded}}^1 \\ \vdots \end{bmatrix} \\ \begin{bmatrix} \vdots \\ \vdots \end{bmatrix} & \begin{bmatrix} \mathbf{S}_{2\text{-noded}}^2 \\ \vdots \end{bmatrix} \\ \vdots & \vdots \\ \begin{bmatrix} \vdots \\ \vdots \end{bmatrix} & \begin{bmatrix} \mathbf{S}_{2\text{-noded}}^{n-1} \\ \mathbf{S}_{2\text{-noded}}^n \end{bmatrix} \end{bmatrix}$$

The shaded portion is not used $\Rightarrow$

(b)

Figure 4. Global Stiffness Matrix Construction for Pavement Structures (a) with a Halfspace and (b) without a Halfspace (i.e., shallow bedrock)

Once the global stiffness matrix is constructed, the displacements at every node of the pavement structure can be found simply by inverting the stiffness matrix. In other words:

$$\hat{\mathbf{U}} = \mathbf{S}_{\text{Global}}^{-1} \cdot \hat{\mathbf{P}} \quad (61)$$

where

$$\hat{\mathbf{U}} = \left\{ \hat{U}_{x,1} \quad \hat{U}_{y,1} \quad \hat{U}_{z,1} \quad \cdots \quad \hat{U}_{x,n-1} \quad \hat{U}_{y,n-1} \quad \hat{U}_{z,n-1} \right\}^T \quad (62)$$

is the global displacement vector, and $\hat{U}_{x,i}$, $\hat{U}_{y,i}$, and $\hat{U}_{z,i}$ are the x, y, and z components of the displacement at the i th node, respectively. Similarly, the global force vector $\hat{\mathbf{P}}$ is given as the following.

$$\hat{\mathbf{P}} = \left\{ \hat{P}_{zx,1} \quad \hat{P}_{yz,1} \quad \hat{P}_{z,1} \quad \cdots \quad \hat{P}_{zx,n-1} \quad \hat{P}_{yz,n-1} \quad \hat{P}_{z,n-1} \right\}^T \quad (63)$$

where $\hat{P}_{zx,i}$ and $\hat{P}_{yz,i}$ are the applied shear stresses in the zx and yz planes, respectively, and $\hat{P}_{z,i}$ is the applied normal stress in the z direction, all of which are at the i th node.

For pavements that are only subjected to the vertical load at the pavement surface, the only non-zero component of $\hat{\mathbf{P}}$ in Equation (63) is the normal component $\hat{P}_{z,1}$, which can be obtained easily by taking the forward Fourier transform of the applied pressure at the pavement surface, $p(x, y)$. Mathematically, this is written as the following.

$$\hat{P}_{z,1}(k_x, k_y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p(x, y) e^{-ik_x x} e^{-ik_y y} dx dy \quad (64)$$

Time-Domain Response

Note that the global displacement vector $\hat{\mathbf{U}}$ obtained from Equation (61) is still in the transformed domain. As discussed previously, the advantage of working in the transformed domain is that the partial derivatives in the physical domain are represented as algebraic multiplications in the transformed domain.

More specifically, because the deflection slope with respect to x at the pavement surface is represented as $\partial U_{z,1} / \partial x$, its frequency domain counterpart is simply obtained as $ik_x \hat{U}_{z,1}$. Both the deflection and deflection slope can then be brought back to the physical domain through the respective 2D inverse Fourier transforms as shown in the following equations.

$$U_{z,1} = \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \hat{U}_{z,1}(k_x, k_y) e^{ik_x x} e^{ik_y y} dk_x dk_y \quad (65)$$

$$\frac{\partial U_{z,1}}{\partial x} = \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} ik_x \hat{U}_{z,1}(k_x, k_y) e^{ik_x x} e^{ik_y y} dk_x dk_y \quad (66)$$

From Formulation to DSteele

The preceding derivation provides the mathematical foundation implemented in DSteele. At the element level, each finite pavement layer is represented by a stiffness matrix in the frequency domain, while the bottom layer can be represented directly as an element of finite thickness or a semi-infinite element. These element relationships are assembled into the global pavement system in a manner analogous to finite-element assembly.

For a specified moving load, the surface traction is represented in the same transformed domain as the pavement system. Solving the global system provides nodal displacements in the frequency domain. The inverse transformation then returns the response back to physical domain, allowing the model to describe the pavement response produced by a load traveling at a prescribed constant speed.

Response Quantities and TSD Applications

Although the formulation is developed in terms of the full three-dimensional displacement and stress fields, DSteele was developed with moving-deflection applications as a central motivation. Vertical surface deflection is obtained naturally from the nodal displacements, and the deflection slope needed for Traffic Speed Deflectometer (TSD) analysis is also obtained easily as its derivative.

Relationship to the DSteele Series

DS-01 provides the development history and engineering overview of DSteele. DS-02 documents the underlying mathematical formulation. A subsequent article will address DSteele V2 and the computational changes used to accelerate the original implementation. Additional articles can then treat TSD measurement and integration, direct deflection-slope backcalculation, dimensional analysis, and the DSteele machine-learning surrogate without repeatedly reproducing the derivation presented here.

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